

SHL

Question

A company manufactures different types of software products. They deliver their products to their N clients. Whenever the company fulfills the complete order of a client, the orderID generated is the concatenation of the number of products delivered for every committed product type. The head of the sales team wishes to find the clientwise data for the total number of products of any type delivered to every client.

Write an algorithm for the head of sales team to calculate the total number of products of any type delivered to the respective clients.

Input

The first line of the input consists of an integer $numOfClients$, representing the number of clients (N). The second line consists of N space-separated integers - $orderID_0, orderID_1, \dots, orderID_{N-1}$, representing the orderIDs of the orders delivered to the clients.

Output

Print N space-separated integers representing the clientwise data for the total number of products of any type delivered to each of the respective clients.

Constraints

$$0 \leq numOfClients \leq 10^6$$

$$0 \leq orderID \leq 10^6$$

$$0 \leq i < numOfClients$$

Example

Input:

4
43 345 20 987

Output:

7 12 2 24

Explanation:

The client 0 ordered 2 types of products; thus, the total products delivered to him/her is 7.

The client 1 ordered 3 types of products; thus, the total products delivered to him/her is 12.

The client 2 ordered 2 types of products; thus, the total products delivered to him/her is 2.

43:56

Show tips

Help

Accessibility

C++ (14)

```
7
8 /*
9 *
10 */
11 vector<int> prodDelivery (vector<int> orderID)
12 {
13     vector<int> answer;
14     // write your code here
15
16     return answer;
17 }
18
19 int main()
20 {
21
22
23     //input for orderID
24     int orderID_size;
25     cin >> orderID_size;
26     vector<int> orderID;
27     for ( int idx = 0; idx < orderID_size; idx++ )
28     {
29         int temp;
30         cin >> temp;
31         orderID.push_back(temp);
32     }
33
34     //output
35     vector<int> result = prodDelivery(orderID);
36     for ( int idx = 0; idx < result.size() - 1; idx++ )
37     {
38         cout << result[idx] << " ";
39     }
40     cout << result[result.size() - 1];
41
42     return 0;
43 }
44
```

Test Cases & Output

SHL

Question

An employee in an organisation has begun working on N projects (numbered 0 to $N-1$). Each week he/she can work on a single module of one of the projects. The modules that are chosen on any two successive weeks should come from different projects. A project i can have at most C_i modules. The modules of the projects are such that a module is completed in a week.

Write an algorithm to determine the number of weeks the employee can work on projects following the above-mentioned rules.

Input

The first line of the input consists of an integer - num , representing the number of projects (N).

The next line consists of N space-separated integers - $projC_0, projC_1, \dots, projC_{N-1}$, representing the number of modules of the projects.

Output

Print an integer representing the maximum number of weeks the employee can work on the projects.

Constraints

$$1 \leq num \leq 5 \times 10^4$$

$$1 \leq projC_i \leq 10^7$$

$$1 \leq projC_i \leq 10^5$$

$$0 \leq i < num$$

Example

Input:

3

Output:

Explanation:

second and third projects have 7, 2 and 3 modules respectively.

Modules of different projects are selected on successive weeks in a sequence: first, second, first, third,

first, third, first, third, first.

Maximum number of weeks an employee can work on these projects is 11.

39:25

Code saved

```
1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5
6
7
8 //
9
10
11 int workingWeeks (vector<int> projC)
12 {
13     int answer;
14     // Write your code here
15
16
17     return answer;
18 }
19
20 int main()
21 {
22
23     //input for projC
24     int projC_size;
25     cin >> projC_size;
26     vector<int> projC;
27     for ( int idx = 0; idx < projC_size; idx++ )
28     {
29         int temp;
30         cin >> temp;
31         projC.push_back(temp);
32     }
33
34     int result = workingWeeks(projC);
35     cout << result;
36
37
38     return 0;
39 }
```

Test Cases & Output

RUN SOLUTION

SHL

PASSAGE

Table Employee:

EMPID	EMPNAME	DEPT	POSITION	SALARY
1	X	C	SE	4000
2	Y	C++	SSE	8000
3	Y	JAVA	TL	10000
4	P	SQL	PM	12000
5	Q	UNIX	SE	4000
6	R	C++	SE	4000

Choose the correct option.

What will be the output of the following query when executed on the given table "Employee"?

```
SELECT EMPID  
FROM Employee  
WHERE POSITION IN (SELECT POSITION FROM Employee WHERE SALARY > 5000);
```

OPTIONS

5 records

6 records

3 records

4 records

SKIP

General Ability

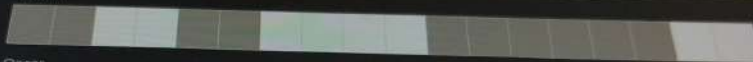
You are encouraged to take the practice test, which includes tips on how to answer the different question types included in the test.

Testing Tips

- Whenever "weekdays" are referenced, assume this means Monday to Friday.
- For the 30-day calendar items, you may select up to 5 dates.
- For pie chart questions, a single wedge cannot be set to be lower than 5%.
- For bar chart questions, the overall bar value cannot be lower than the lowest value on the Y-axis.
- **For the best experience with this assessment, we recommend that you use the most up-to-date version of your browser. We recommend using Chrome, Firefox, Safari, or Edge as your browser.**
- Please try to work quickly but accurately. Do not spend too long on any one question. **You must complete the entire test to avoid a score reduction.**
- Read the provided information carefully before responding to each question.
- It is recommended you complete the entire test in one sitting.
- If you lose your connection, the test will restart from where you left it.
- You will not have an opportunity to review your answers at the end of this test.
- Please note: If completing this test on a mobile device, we recommend you turn off all notifications. The test timer will not pause if you take a call or respond to text messages while in the test.

Team Calendar

Naomi



Oscar



Petra



Question

Reschedule the Team Meeting, currently set for 3:30 p.m., to an earlier time. You may move one appointment per person, but not more than 30 minutes earlier or later. Your appointments and client demos cannot be moved.

Your Calendar



Next

Take Candidate Assessment x General Ability | TalentCentr... x

Sunday May 24, 2:43:13 PM

talentcentral.eu.shl.com/show/TestItem/branches=01174154-F09C-4029-AD9-3-06ACDF3C2-testItem/...

NatWest Group

Team Calendar

Naomi

Oscar

Petra

9 10 11 12 1 2 3 4 5 6

Question

Reschedule the Team Meeting, currently set for 3:30 p.m., to an earlier time. You may move one appointment per person, but not more than 30 minutes earlier or later. Your appointments and client demos cannot be moved.

Your Calendar

9 10 11 12 1 2 3 4 5 6

Next

Team Calendar

Naomi



Oscar



Petra



Feedback

You have partially answered the question correctly. Please either attempt the question again or continue.

Try Again

Continue

Question

Reschedule the Team Meeting, currently set for 3:30 p.m., to an earlier time. You may move one appointment per person, but not more than 30 minutes earlier or later. Your appointments and client demos cannot be moved.

Your Calendar



Next

Team Calendar

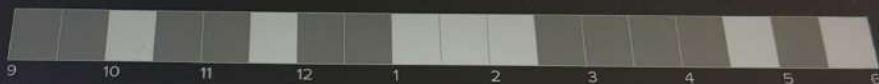
Naomi



Oscar



Petra



9 10 11 12 1 2 3 4 5 6

Question

Reschedule the Team Meeting, currently set for 3:30 p.m., to an earlier time. You may move one appointment per person, but not more than 30 minutes earlier or later. Your appointments and client demos cannot be moved.

Your Calendar



9 10 11 12 1 2 3 4 5 6

Feedback

You have answered the question correctly. Please continue.

Try Again

Continue

Next

Question

Schedule the tasks as early as possible. Note that your laptop is being repaired today. Two tasks (Data Entry and Run Reports) cannot be done before you pick up your laptop.



Today



Data Entry 1 hr

Sales Meeting 1 hr

Run Reports 1.5 hrs

Interview 1.5 hrs

[Next](#)

Schedule the tasks as early as possible. Note that your laptop is being repaired today. Two tasks (Data Entry and Run Reports) cannot be done before you pick up your laptop.



Meeting Room Requirements

	Fei	Any room – 4-hour meeting
	George	East room – 2-hour meeting
	Helen	Any room – 6-hour meeting
	Ian	Any room – 3-hour meeting

Question

Rooms are available from 8:00 a.m. to 4:00 p.m. Match each person's meeting to a room so that they can all be scheduled today.



	West	East



Travel Meal Allowance

- Employee Only Meals: Breakfast \$12 per person; Lunch \$15 per person; Dinner \$28 per person.
- Meals with Clients: Breakfast \$20 per person; Lunch \$30 per person; Dinner \$56 per person.
- Total meals expensed must not exceed allowed expenses based on rates given.

Question

Decide if total meal reimbursement requests should be approved.



Meal	# of People	Cost
Employee Breakfast	2	\$26.53
Client Lunch	3	\$75.45
Client Dinner	3	\$150.04
Total		\$252.02
Approval Status		
Andrew	Bella	Connor

Approved

Not Approved

Next

- Employee Only Meals: Breakfast \$12 per person; Lunch \$15 per person; Dinner \$28 per person.
- Meals with Clients: Breakfast \$20 per person; Lunch \$30 per person; Dinner \$56 per person.
- Total meals expensed must not exceed allowed expenses based on rates given.

Decide if total meal reimbursement requests should be approved.



Approved ☐ Not Approved ☐

```
er/launch?s=D1174454-F00C-4D39-ADF3-684CDC8CF662&fromLaunch=true#tabpane1
```

Travel Meal Allowance

- Employee Only Meals: Breakfast \$12 per person; Lunch \$15 per person; Dinner \$28 per person;
- Meals with Clients: Breakfast \$20 per person; Lunch \$30 per person; Dinner \$56 per person;
- Total meals expensed must not exceed allowed expenses based on rates given.

Question

Decide if total meal reimbursement requests should be approved.



Meal	# of People	Cost
Client Breakfast	6	\$119.78
Client Lunch	5	\$137.87
Dinner	--	--
Total		\$257.65
Approval Status		
Andrew	Bella	Connor

Approved

Not Approved

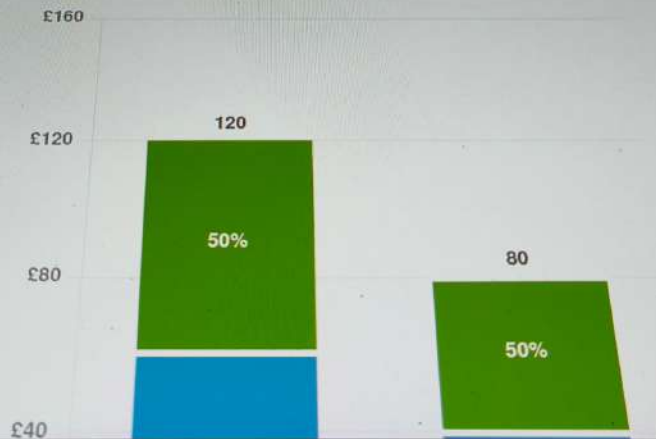
Next

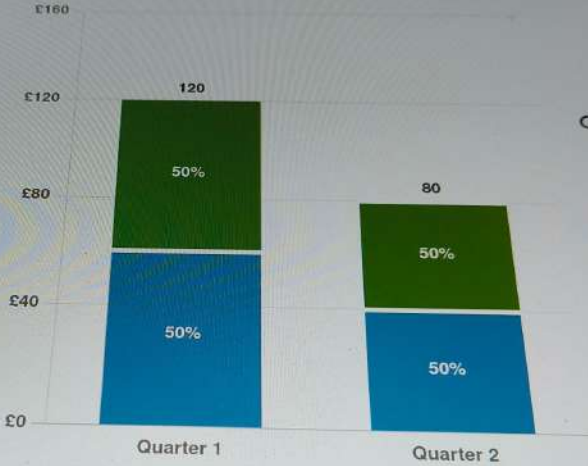
Sales Revenue

- Total sales revenue was £120 million in Quarter 1.
- Total sales revenue in Quarter 2 decreased by £10 million compared to Quarter 1.
- Sales revenue in the East was £78 million in Quarter 1.
- Sales revenue in the West increased by £46 million from Quarter 1 to Quarter 2.

Question

Arrange the sales revenue (in '000,000s) by quarter and the percentages in the **East** and **West** regions.





Next

Loyalty Discount

- 5% discount: Average monthly spending of £20 or more and contract length of 6 months or longer.
- 10% discount: Average monthly spending of £30 or more and contract length of 12 months or longer.

Question

Work out the correct discount for each order.



Total spending	£278			
Contract length	7 months			
Discount				
	Order 1	Order 2	Order 3	Order 4

- 0%
- 5%
- 10%

Next

Loyalty Discount

- 5% discount: Average monthly spending of £20 or more and contract length of 6 months or longer.
- 10% discount: Average monthly spending of £30 or more and contract length of 12 months or longer.

Question

Work out the correct discount for each order.



Total spending	£383			
Contract length	2 years			
Discount				
Order 1	Order 2	Order 3	Order 4	

0%

5%

10%

Next



Take Candidate Assessment

General Ability | Tabpane2

35:44

Loyalty Discount

- 5% discount: Average monthly spending of £20 or more and contract length of 6 months or longer.
- 10% discount: Average monthly spending of £30 or more and contract length of 12 months or longer.

Question

Work out the correct discount for each order.

Total spending	£589			
Contract length	1 year 8 months			
Discount				
	Order 1	Order 2	Order 3	Order 4

0%

5%

10%

Next

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entcentral.eu.shl.com/player/testdriver/launch?s=ABB61048-307F-4DA0-BEF1-4AC6374F3951&fromLaunch=true#tabpane2



35:41

Loyalty Discount

- 5% discount: Average monthly spending of £20 or more and contract length of 6 months or longer.
- 10% discount: Average monthly spending of £30 or more and contract length of 12 months or longer.

Question

Work out the correct discount for each order.



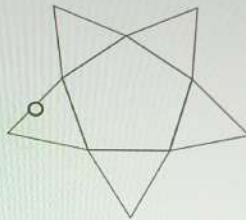
Total spending	£472		
Contract length	1 year 3 months		
Discount			
	Order 1	Order 2	Order 3
			Order 4
	0%	5%	10%

Next

33:50

Question

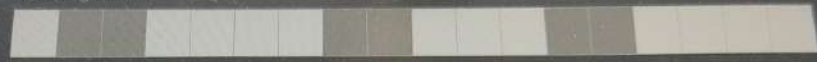
Tap to fill the next image in the sequence



Next

Editing Schedule

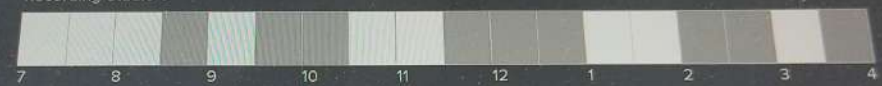
Recording Studio 1



Recording Studio 2



Recording Studio 3



Question

Someone has reserved Recording Studio 3 from 8:00 a.m. - 10:00 a.m., and all the reservations in Recording Studio 1 have been pushed back an hour. Keeping in mind that you can cancel 30 minutes of one reservation, find a time to shut down all Recording Studios for one hour for maintenance.

Recording Studio 4



Next

Sales by Quarter
A company made sales of \$1,254,000 last year.

Quarter 1

Quarter 2

Produced $\frac{1}{3}$ more sales than in quarter 1

Quarter 3

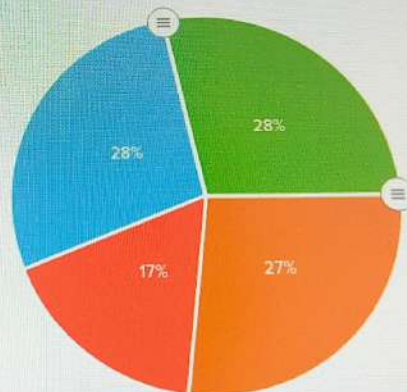
Produced 17% of total sales for the year

Quarter 4

Sales increased 100% over the previous quarter

Question

Adjust the pie chart to represent the sales each quarter.



Sales by Quarter

A company made sales of \$1,254,000 last year.

Quarter 1

Quarter 2

Produced 1/3 more sales than in quarter 1

Quarter 3

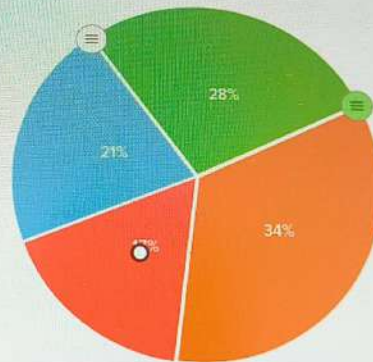
Produced 17% of total sales for the year

Quarter 4

Sales increased 100% over the previous quarter

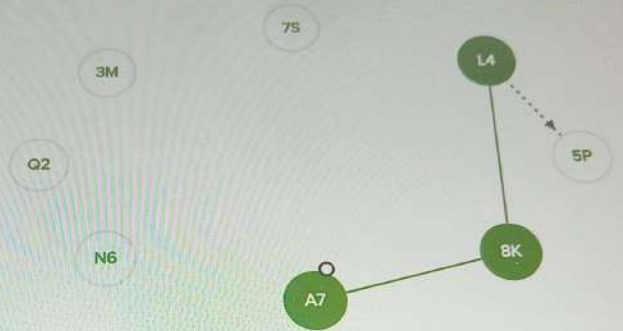
Question

Adjust the pie chart to represent the sales each quarter.



Question

Drag the line to finish the sequence.



Next

Mandatory Question(s)

You have not answered the question. Please answer the question in full in order to proceed.

[Continue](#)

Q2

N6

A7

8K

5P

[Next](#)

	Research	✗ Can't be first
	Finance	✓ Must be immediately before Legal
	Sales	✓ Available immediately before or after Finance
	Marketing	✓ Must be completed in the first half of the month
	Legal	✗ Can't be last

Question

Each department is holding a leadership team meeting for four consecutive business days. Meetings cannot overlap. What are the start and end dates for the Research meeting?

Calendar

SUN	MON	TUE	WED	THU	FRI	SAT
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

Information

There are 110,000 litres of petrol in total available in a chain of four petrol stations.

Station A
Three times as much as Station C

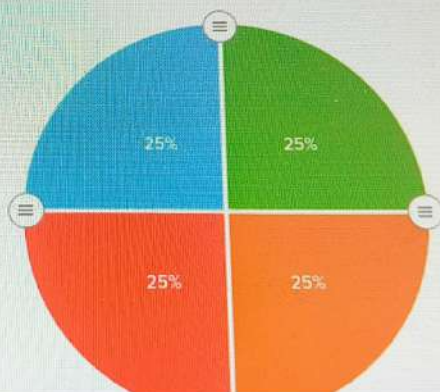
Station B
24,200 litres more than Station A

Station C

Station D
24,200 litres more than Station C

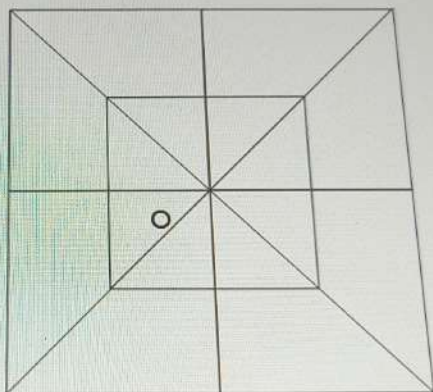
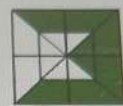
Question

Adjust the pie chart to represent the amount of petrol at each station (rounding to the nearest whole number only in the last step).



Question

Tap to fill the next image in the sequence



Next

Question

Tap to fill the next image in this sequence

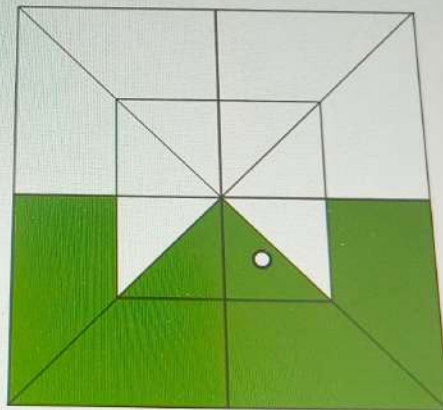


Next

21:18

Question

Tap to fill the next image in the sequence



Next

	Human Resources	Submitted the first week of the month
	Research	Submitted two weeks before Sales
	Sales	Submitted before the last week of the month
	Marketing	Submitted in the later half of the month
	Facilities	Submitted after Research but before Marketing

Question

Five departments submitted monthly expense reports on different Thursdays last month. Finance then spent three weekdays (starting Friday) to process each report. On which three days was Finance processing the expense report from Facilities?

Calendar

SUN	MON	TUE	WED	THU	FRI	SAT
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25

20.00

Stationery

- Pens cost £2.20 each.
- Pencils cost £0.80 each.
- Exactly £23 was spent on a combined total of 20 pens and pencils.

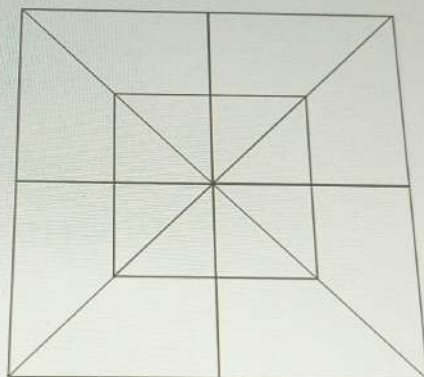
Question

Arrange the numbers to show what percentage of items purchased were **pens** and **pencils**.



Question

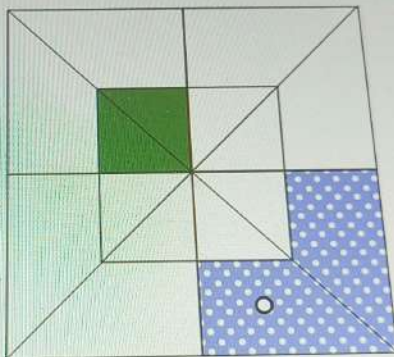
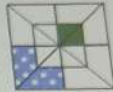
Tap to fill the next image in the sequence



Next

Question

Tap to fill the next image in the sequence



Next

Work Schedule

16:02

	Isabel	✓	Available 1st-8th and 14th-21st
	Jack	✓	Works Mondays, Wednesdays, Fridays
	Katarina	✗	Does not work Mondays, Fridays
	Liam	✓	Available 25th-29th

Question

At least two employees must be scheduled to work each weekday. Select all the weekdays that need additional employees scheduled.

Calendar

SUN	MON	TUE	WED	THU	FRI	SAT
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

15:18

Student Enrolment

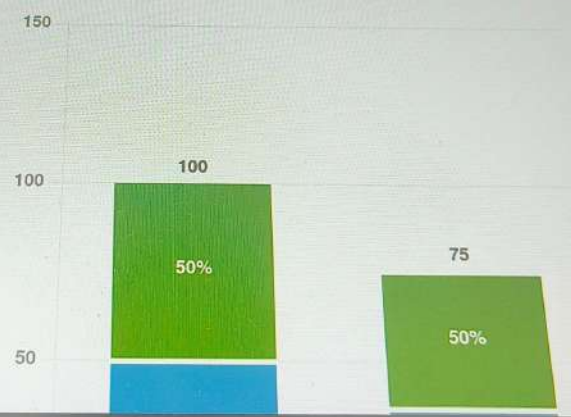
- During Semester 1, the male to female student ratio in a class was 3:1.
- Male student enrolment in Semester 2 is 20% higher than in Semester 1.
- Female student enrolment in Semester 2 is 40% higher than in Semester 1.
- There were a total of 100 students during Semester 1.

Question

Arrange the number of students enrolled for each semester and the **male** and **female** percentages.



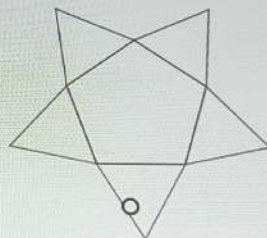
O



14:08

Question

Tap to fill the next image in the sequence



Next

13:34

Typing Speed



Charlie and Daniela have exactly two employees between them.



Daniela



Ian is faster than Quentin with exactly one employee between them.



Jasmine



Quentin is faster than Daniela with exactly one employee between them.



Thandi is slower than Daniela.

Question

Drag the badges to each employee, sorting them from 1 (fastest) to 6 (slowest).



○



Charlie

#



Daniela

#



Ian

#



Jasmine

#



Quentin

#



Thandi

#

Badges

1

2

3

4

5

6

Information

Information on the annual revenues of four business units is provided below.

Americas

Half the revenue of Asia-Pacific

Europe

£562,500

Asia-Pacific

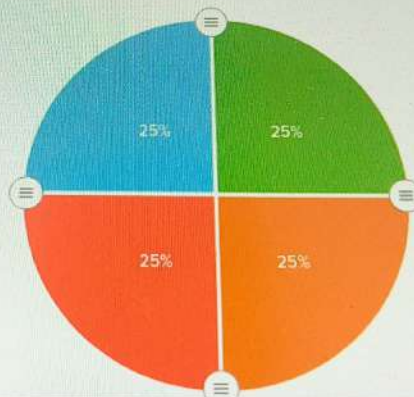
80% of the revenue of Europe

Africa

Ratio of 9:5 with Europe

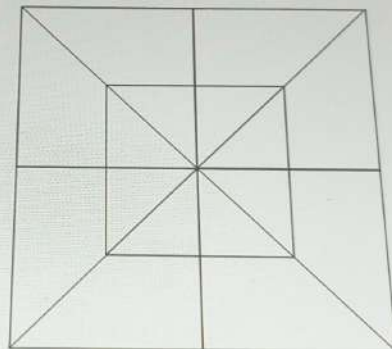
Question

Adjust the pie chart to represent the percentage of total revenue accounted for by each business unit.



Question

Tap to fill the next image in the sequence



Next

Team Calendar

Harry



Isabel



Jack



Question

Harry spends 1:00 - 2:00 p.m. each day running reports. You need 30 minutes to prepare for a strategic planning meeting with your team. Find a time to schedule the one-hour strategic meeting.

Your Calendar



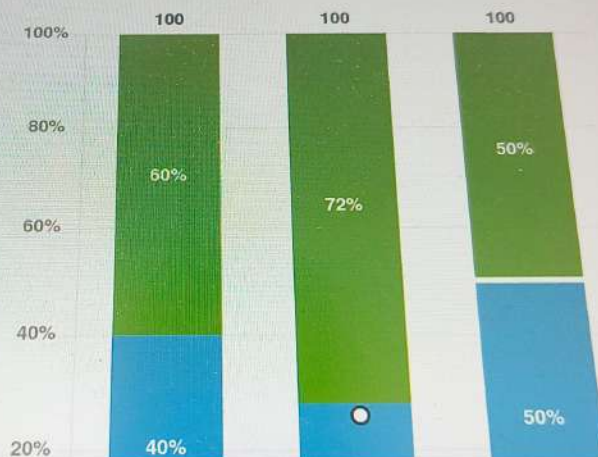
Next

Contract Renewals

- The percentage of renewed contracts was 60% for Year 1 and 72% for Year 2.
- The renewed contract percentage increase from Year 2 to Year 3 is equal to $\frac{3}{4}$ of the percentage point increase from Year 1 to Year 2.

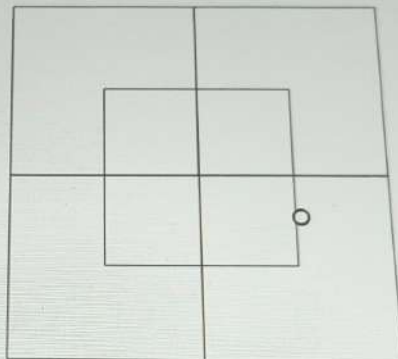
Question

Arrange the **cancelled contracts** and **renewed contracts** percentages for Year 3.



Question

Tap to fill the next image in the sequence



Next

Installation Schedule

	Building A	✗ Can't happen first
	Building B	✓ Must happen before Building D
	Building C	✓ Must include the 12th in the schedule
	Building D	✓ Must happen before Building A

Question

New security systems must be installed on four buildings, one at a time. Installation takes four days. The installers do not start a new job on Mondays or work weekends. On which date should the work on Building A begin, to complete all four buildings as early as possible?



Calendar

SUN	MON	TUE	WED	THU	FRI	SAT
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27

Company Expenses

A small business has €400,000 in expenses.

Rent

When combined with Utilities, is the same as Wages

Utilities

Cost €24,000

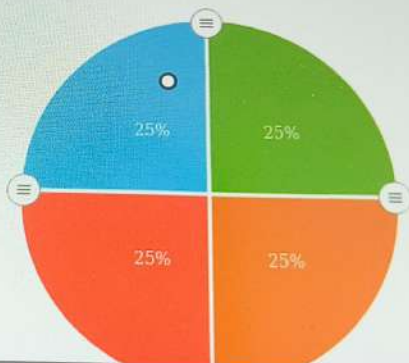
Taxes

Cost €20,000 less than wages

Wages

Question

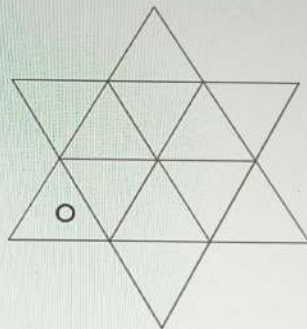
Adjust the pie chart to represent the expenses.



04:53

Question

Tap to fill the next image in the sequence

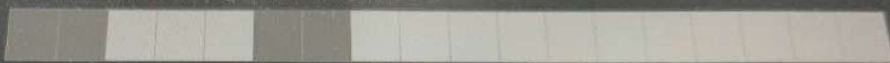


Next

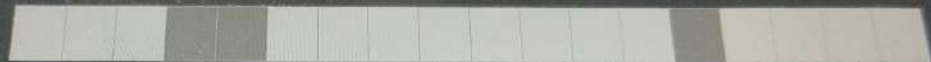
Machine Schedule

03:52

Machine 1: can't run at the same time as Machine 4.



Machine 2: can run at the same time as Machine 4.



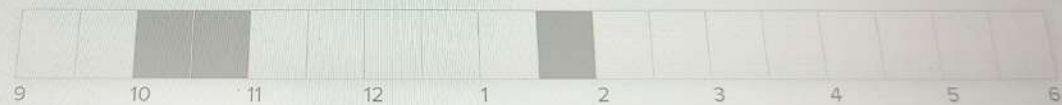
Machine 3: can't run at the same time as Machine 4.



Question

The coloured blocks represent times when the machines are scheduled to run. A 60-minute gap is needed after running Machine 3 before starting Machine 4. Schedule a one-hour time for Machine 4.

Machine 4



Next